

P8131 Homework 5

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2026-03-24

Problem 1: Horseshoe Crabs

In a study of nesting horseshoe crabs, each female horseshoe crab had a male crab attached to her in her nest. The study investigated factors that affect whether the female crab had any other males, called satellites, residing near her. Explanatory variables that are thought to affect this included the female crab's color (C), spine condition (S), carapace width (W) and weight (Wt). The response outcome for each female crab is her number of satellites (Sa). There are 173 females in this study. Data are provided in the crab.txt.

```
crab <- read.table("HW5-crab.txt", header = TRUE)
```

(a) Poisson Model M1 with W

Fit a Poisson model (M1) with log link with W as the single predictor. Check the goodness of fit and interpret your model.

Model: $\log(\mu_i) = \beta_0 + \beta_1 W_i$, where $Y_i \sim \text{Poisson}(\mu_i)$.

```
M1 <- glm(Sa ~ W, family = poisson(link = "log"), data = crab)
summary(M1)
```

```
##
## Call:
## glm(formula = Sa ~ W, family = poisson(link = "log"), data = crab)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -3.30476    0.54224  -6.095  1.1e-09 ***
## W           0.16405    0.01997   8.216  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 632.79  on 172  degrees of freedom
## Residual deviance: 567.88  on 171  degrees of freedom
## AIC: 927.18
##
## Number of Fisher Scoring iterations: 6
```

Goodness of fit:

```
dev_M1 <- deviance(M1)
df_M1 <- M1$df.residual
p_dev_M1 <- 1 - pchisq(dev_M1, df_M1)

G_M1 <- sum(residuals(M1, type = "pearson")^2)
p_G_M1 <- 1 - pchisq(G_M1, df_M1)

cat("Deviance =", round(dev_M1, 2), ", df =", df_M1,
    ", p =", round(p_dev_M1, 4), "\n")
```

```
## Deviance = 567.88 , df = 171 , p = 0
```

```
cat("Pearson  $\chi^2$  =", round(G_M1, 2), ", df =", df_M1,
    ", p =", round(p_G_M1, 4), "\n")
```

```
## Pearson  $\chi^2$  = 544.16 , df = 171 , p = 0
```

Both the deviance and Pearson χ^2 statistics are large relative to their degrees of freedom ($p < 0.05$), indicating that the model does not fit the data well. There is evidence of lack of fit, possibly due to over-dispersion or missing predictors.

Interpretation: The estimated model is $\log(\hat{\mu}) = -3.3048 + 0.164 \cdot W$.

$\exp(\hat{\beta}_1)$ is the estimated rate ratio for the number of satellites associated with a one-cm increase in carapace width. With $\hat{\beta}_1 = 0.164$ ($p < 0.001$), the expected number of satellites for a female crab is 1.178 ($= \exp(0.164)$) times of that for a crab one cm narrower in width.

(b) Model M2 with W and Wt

Fit a model (M2) with W and Wt as predictors. Compare it with the model in (a). Interpret your results.

```
M2 <- glm(Sa ~ W + Wt, family = poisson(link = "log"), data = crab)
summary(M2)
```

```
##
## Call:
## glm(formula = Sa ~ W + Wt, family = poisson(link = "log"), data = crab)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.29168    0.89929  -1.436  0.15091
## W           0.04590    0.04677   0.981  0.32640
## Wt          0.44744    0.15864   2.820  0.00479 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 632.79  on 172  degrees of freedom
```

```
## Residual deviance: 559.89  on 170  degrees of freedom
## AIC: 921.18
##
## Number of Fisher Scoring iterations: 6
```

Model comparison via deviance analysis (LRT):

```
test_stat <- deviance(M1) - deviance(M2)
df_test <- df_M1 - M2$df.residual
p_test <- 1 - pchisq(test_stat, df_test)

cat("Delta Deviance =", round(test_stat, 3), ", df =", df_test,
    ", p =", round(p_test, 4), "\n")
```

```
## Delta Deviance = 7.993 , df = 1 , p = 0.0047
```

The test statistic is 7.993 on 1 df with $p = 0.0047 < 0.05$, so adding Wt leads to a statistically significant improvement. However, the deviance only decreases from 567.88 to 559.89, both far exceeding their respective degrees of freedom, indicating that neither M1 nor M2 fits the data adequately (likely due to over-dispersion).

Interpretation: In M2, W is no longer significant ($p = 0.326$) while Wt is significant ($p = 0.005$). This suggests that weight captures most of the information about satellite count that width provided, and the two predictors are likely correlated.

$\exp(\hat{\beta}_{Wt})$ is the estimated rate ratio for the number of satellites associated with a one-kg increase in weight, holding width fixed. With $\hat{\beta}_{Wt} = 0.4474$, the expected number of satellites is 1.564 ($= \exp(0.4474)$) times of that for a crab one kg lighter, holding width constant.

$\exp(\hat{\beta}_W)$ is the estimated rate ratio for a one-cm increase in width, holding weight fixed. With $\hat{\beta}_W = 0.0459$, W is not significant ($p = 0.326$), suggesting width does not provide additional information beyond weight.

(c) Over-dispersion in M2

Check over dispersion in M2. Interpret the model after adjusting for over dispersion.

```
G_M2 <- sum(residuals(M2, type = "pearson")^2)
phi_hat <- G_M2 / M2$df.residual

cat("Pearson  $X^2$  =", round(G_M2, 2), "\n")
```

```
## Pearson  $X^2$  = 536.6
```

```
cat("Estimated dispersion parameter: phi =", round(phi_hat, 4), "\n")
```

```
## Estimated dispersion parameter: phi = 3.1564
```

Since $\hat{\phi} = 3.16$ is substantially greater than 1, there is evidence of over-dispersion.

Adjusted model using quasi-Poisson:

```
M2_quasi <- glm(Sa ~ W + Wt, family = quasipoisson(link = "log"),
               data = crab)
summary(M2_quasi)
```

```
##
## Call:
## glm(formula = Sa ~ W + Wt, family = quasipoisson(link = "log"),
##      data = crab)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.29168    1.59771  -0.808   0.420
## W             0.04590    0.08309   0.552   0.581
## Wt            0.44744    0.28184   1.588   0.114
##
## (Dispersion parameter for quasipoisson family taken to be 3.156449)
##
## Null deviance: 632.79  on 172  degrees of freedom
## Residual deviance: 559.89  on 170  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 6
```

The updated model has the same coefficient estimates as M2. After adjusting for over-dispersion, the standard errors are inflated by a factor of $\sqrt{\hat{\phi}} = 1.777$. In the adjusted model, neither W ($p = 0.581$) nor Wt ($p = 0.114$) is individually significant at $\alpha = 0.05$.

Negative binomial regression as a comparison:

```
library(MASS)
M2_nb <- glm.nb(Sa ~ W + Wt, data = crab)
summary(M2_nb)
```

```
##
## Call:
## glm.nb(formula = Sa ~ W + Wt, data = crab, init.theta = 0.9323857725,
##         link = log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.38857    1.82569  -0.761   0.4469
## W             0.02889    0.09682   0.298   0.7654
## Wt            0.66340    0.35182   1.886   0.0593 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(0.9324) family taken to be 1)
##
## Null deviance: 216.60  on 172  degrees of freedom
## Residual deviance: 196.24  on 170  degrees of freedom
## AIC: 756.57
##
```

```
## Number of Fisher Scoring iterations: 1
##
##
##           Theta:  0.932
##         Std. Err.:  0.168
##
##  2 x log-likelihood: -748.572
```

The NB model assumes $\text{Var}(Y) = \mu + \mu^2/\theta$. The estimated $\hat{\theta} = 0.93$ with a finite standard error, confirming over-dispersion. The coefficient estimates are similar to those from quasi-Poisson, and neither W nor Wt is individually significant, consistent with the quasi-Poisson results.

Problem 2: Fish Parasites

Researchers examined a large number of fish to determine the prevalence of parasites. The dataset (parasite.txt) includes the variables Intensity (i.e., the number of parasites), Area (a categorical variable), Year (to be treated as categorical), and Length of the fish.

```
parasite <- read.table("HW5-parasite.txt", header = TRUE, sep = "\t")
parasite <- parasite %>%
  dplyr::select(Sample, Intensity, Year, Length, Area)
parasite$Area <- factor(parasite$Area)
parasite$Year <- factor(parasite$Year)
```

(a) Poisson Model

Fit a Poisson model with log link to the data with area, year, and length as predictors. Interpret each model parameter.

Model: $\log(\mu_i) = \beta_0 + \beta_1 \text{Area}_2 + \beta_2 \text{Area}_3 + \beta_3 \text{Area}_4 + \beta_4 \text{Year}_{2000} + \beta_5 \text{Year}_{2001} + \beta_6 \text{Length}$

```
fit_pois <- glm(Intensity ~ Area + Year + Length,
  family = poisson(link = "log"), data = parasite)
summary(fit_pois)
```

```
##
## Call:
## glm(formula = Intensity ~ Area + Year + Length, family = poisson(link = "log"),
##      data = parasite)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  2.6431709  0.0542838  48.692  < 2e-16 ***
## Area2        -0.2119557  0.0491691  -4.311  1.63e-05 ***
## Area3        -0.1168602  0.0428296  -2.728  0.00636 **
## Area4         1.4049366  0.0356625  39.395  < 2e-16 ***
## Year2000      0.6702801  0.0279823  23.954  < 2e-16 ***
## Year2001     -0.2181393  0.0287535  -7.587  3.29e-14 ***
## Length       -0.0284228  0.0008809 -32.265  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 25797  on 1190  degrees of freedom
## Residual deviance: 19153  on 1184  degrees of freedom
##      (63 observations deleted due to missingness)
## AIC: 21089
##
## Number of Fisher Scoring iterations: 7
```

Interpretation of parameters:

```
coefs <- round(coef(fit_pois), 4)
```

$\exp(\hat{\beta}_0) = 14.06$ ($= \exp(2.6432)$) is the estimated expected parasite count for a fish in Area 1, Year 1999, with Length = 0.

$\exp(\hat{\beta}_1)$ is the estimated rate ratio comparing Area 2 to Area 1, holding year and length fixed. With $\hat{\beta}_1 = -0.212$, the expected parasite count for fish in Area 2 is 0.809 ($= \exp(-0.212)$) times of that in Area 1.

$\exp(\hat{\beta}_2)$ is the estimated rate ratio comparing Area 3 to Area 1, holding year and length fixed. With $\hat{\beta}_2 = -0.1169$, the expected parasite count for fish in Area 3 is 0.89 ($= \exp(-0.1169)$) times of that in Area 1.

$\exp(\hat{\beta}_3)$ is the estimated rate ratio comparing Area 4 to Area 1, holding year and length fixed. With $\hat{\beta}_3 = 1.4049$, the expected parasite count for fish in Area 4 is 4.075 ($= \exp(1.4049)$) times of that in Area 1.

$\exp(\hat{\beta}_4)$ is the estimated rate ratio comparing Year 2000 to Year 1999, holding area and length fixed. With $\hat{\beta}_4 = 0.6703$, the expected parasite count in 2000 is 1.955 ($= \exp(0.6703)$) times of that in 1999.

$\exp(\hat{\beta}_5)$ is the estimated rate ratio comparing Year 2001 to Year 1999, holding area and length fixed. With $\hat{\beta}_5 = -0.2181$, the expected parasite count in 2001 is 0.804 ($= \exp(-0.2181)$) times of that in 1999.

$\exp(\hat{\beta}_6)$ is the estimated rate ratio for a one-unit increase in fish length, holding area and year fixed. With $\hat{\beta}_6 = -0.0284$, the expected parasite count is 0.972 ($= \exp(-0.0284)$) times of that for a fish one unit shorter.

(b) Goodness of Fit

Test for goodness of fit of the model in (a) and state conclusions.

```
dev_pois <- deviance(fit_pois)
df_pois <- fit_pois$df.residual
p_dev_pois <- 1 - pchisq(dev_pois, df_pois)

G_pois <- sum(residuals(fit_pois, type = "pearson")^2)
p_G_pois <- 1 - pchisq(G_pois, df_pois)

cat("Deviance =", round(dev_pois, 2), ", df =", df_pois,
    ", p =", round(p_dev_pois, 4), "\n")
```

```
## Deviance = 19152.8 , df = 1184 , p = 0
```

```
cat("Pearson X^2 =", round(G_pois, 2), ", df =", df_pois,
    ", p =", round(p_G_pois, 4), "\n")
```

```
## Pearson X^2 = 42164.97 , df = 1184 , p = 0
```

Both the deviance and Pearson χ^2 are extremely large relative to the degrees of freedom ($p \approx 0$), indicating very poor fit. The Poisson model does not adequately describe these data. This could be due to over-dispersion, excess zeros, or both.

(c) Zero-Inflated Poisson Model

Researchers suspect that there may be two strains of fish, one that is susceptible to parasites and one that is not. Without knowing which fish are susceptible, this could be regarded as a zero-inflated model. Building on the model in (a) (using the same predictors), fit an appropriate model to the data that can account for extra zeros. Provide an interpretation for each model parameter in terms of the problem.

The Zero-Inflated Poisson (ZIP) model assumes two components. Let Z_i be a latent binary variable indicating whether fish i is susceptible ($Z_i = 1$) or non-susceptible ($Z_i = 0$), with $P(Z_i = 0) = \pi_i$. Then:

- With probability π_i , $Y_i | (Z_i = 0) = 0$ (structural zero: non-susceptible fish always have 0 parasites).
- With probability $1 - \pi_i$, $Y_i | (Z_i = 1) \sim \text{Poisson}(\lambda_i)$.

Thus $P(Y_i = 0) = \pi_i + (1 - \pi_i)e^{-\lambda_i}$, and for $y > 0$, $P(Y_i = y) = (1 - \pi_i) \frac{e^{-\lambda_i} \lambda_i^y}{y!}$.

We model both components using the same predictors $\mathbf{x}_i = (1, \text{Area}_i, \text{Year}_i, \text{Length}_i)^T$: $\log(\lambda_i) = \mathbf{x}_i^T \beta$ (Poisson count component) and $\text{logit}(\pi_i) = \mathbf{x}_i^T \gamma$ (zero-inflation component).

```
fit_zip <- zeroinfl(Intensity ~ Area + Year + Length | Area + Year + Length,
                   data = parasite)
summary(fit_zip)
```

```
##
## Call:
## zeroinfl(formula = Intensity ~ Area + Year + Length | Area + Year + Length,
##          data = parasite)
##
## Pearson residuals:
##      Min       1Q   Median       3Q      Max
## -2.1278 -0.8265 -0.5829 -0.1821 25.4837
##
## Count model coefficients (poisson with log link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.8431714  0.0583793  65.831  < 2e-16 ***
## Area2        0.2687835  0.0500467   5.371 7.85e-08 ***
## Area3        0.1463173  0.0439485   3.329 0.000871 ***
## Area4        0.9448068  0.0368342  25.650 < 2e-16 ***
## Year2000     0.3919831  0.0282952  13.853 < 2e-16 ***
## Year2001    -0.0448455  0.0296057  -1.515 0.129833
## Length      -0.0368067  0.0009747 -37.762 < 2e-16 ***
##
## Zero-inflation model coefficients (binomial with logit link):
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.552585  0.275762  2.004 0.04509 *
## Area2        0.718676  0.189552  3.791 0.00015 ***
## Area3        0.657708  0.167402  3.929 8.53e-05 ***
## Area4       -1.022868  0.188201 -5.435 5.48e-08 ***
## Year2000    -0.752119  0.172965 -4.348 1.37e-05 ***
## Year2001     0.456535  0.143962  3.171 0.00152 **
## Length      -0.009889  0.004629 -2.136 0.03266 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Number of iterations in BFGS optimization: 22
## Log-likelihood: -6950 on 14 Df
```

Interpretation of parameters:

Count model (Poisson with log link) — for susceptible fish:

```
coefs_c <- round(coef(fit_zip, "count"), 4)
coefs_z <- round(coef(fit_zip, "zero"), 4)
```

$\exp(\hat{\beta}_0) = 46.67 (= \exp(3.8432))$ is the estimated expected parasite count for a susceptible fish in Area 1, Year 1999, at Length = 0.

$\exp(\hat{\beta}_{\text{Area2}})$ is the estimated rate ratio comparing Area 2 to Area 1 among susceptible fish, holding year and length fixed. With $\hat{\beta}_{\text{Area2}} = 0.2688$, the expected parasite count in Area 2 is 1.308 ($= \exp(0.2688)$) times of that in Area 1.

$\exp(\hat{\beta}_{\text{Area3}})$ is the estimated rate ratio comparing Area 3 to Area 1 among susceptible fish, holding year and length fixed. With $\hat{\beta}_{\text{Area3}} = 0.1463$, the expected parasite count in Area 3 is 1.158 ($= \exp(0.1463)$) times of that in Area 1.

$\exp(\hat{\beta}_{\text{Area4}})$ is the estimated rate ratio comparing Area 4 to Area 1 among susceptible fish, holding year and length fixed. With $\hat{\beta}_{\text{Area4}} = 0.9448$, the expected parasite count in Area 4 is 2.572 ($= \exp(0.9448)$) times of that in Area 1.

$\exp(\hat{\beta}_{\text{Year2000}})$ is the estimated rate ratio comparing Year 2000 to 1999 among susceptible fish, holding area and length fixed. With $\hat{\beta}_{\text{Year2000}} = 0.392$, the expected parasite count in 2000 is 1.48 ($= \exp(0.392)$) times of that in 1999.

$\exp(\hat{\beta}_{\text{Year2001}})$ is the estimated rate ratio comparing Year 2001 to 1999 among susceptible fish, holding area and length fixed. With $\hat{\beta}_{\text{Year2001}} = -0.0448$, the expected parasite count in 2001 is 0.956 ($= \exp(-0.0448)$) times of that in 1999.

$\exp(\hat{\beta}_{\text{Length}})$ is the estimated rate ratio for a one-unit increase in length among susceptible fish, holding area and year fixed. With $\hat{\beta}_{\text{Length}} = -0.0368$, the expected parasite count is 0.9639 ($= \exp(-0.0368)$) times of that for a fish one unit shorter.

Zero-inflation model (logistic with logit link) — probability of being non-susceptible:

$\exp(\hat{\gamma}_0)/(1 + \exp(\hat{\gamma}_0)) = 0.635$ is the estimated probability of being non-susceptible for a fish in Area 1, Year 1999, at Length = 0.

$\exp(\hat{\gamma}_{\text{Area2}})$ is the estimated odds ratio of being non-susceptible comparing Area 2 to Area 1, holding year and length fixed. With $\hat{\gamma}_{\text{Area2}} = 0.7187$, the odds of being non-susceptible in Area 2 is 2.052 ($= \exp(0.7187)$) times of that in Area 1.

$\exp(\hat{\gamma}_{\text{Area3}})$ is the estimated odds ratio of being non-susceptible comparing Area 3 to Area 1, holding year and length fixed. With $\hat{\gamma}_{\text{Area3}} = 0.6577$, the odds of being non-susceptible in Area 3 is 1.93 ($= \exp(0.6577)$) times of that in Area 1.

$\exp(\hat{\gamma}_{\text{Area4}})$ is the estimated odds ratio of being non-susceptible comparing Area 4 to Area 1, holding year and length fixed. With $\hat{\gamma}_{\text{Area4}} = -1.0229$, the odds of being non-susceptible in Area 4 is 0.36 ($= \exp(-1.0229)$) times of that in Area 1.

$\exp(\hat{\gamma}_{\text{Year2000}})$ is the estimated odds ratio of being non-susceptible comparing Year 2000 to 1999, holding area and length fixed. With $\hat{\gamma}_{\text{Year2000}} = -0.7521$, the odds of being non-susceptible in 2000 is 0.471 ($= \exp(-0.7521)$) times of that in 1999.

$\exp(\hat{\gamma}_{\text{Year2001}})$ is the estimated odds ratio of being non-susceptible comparing Year 2001 to 1999, holding area and length fixed. With $\hat{\gamma}_{\text{Year2001}} = 0.4565$, the odds of being non-susceptible in 2001 is 1.579 ($= \exp(0.4565)$) times of that in 1999.

$\exp(\hat{\gamma}_{\text{Length}})$ is the estimated odds ratio of being non-susceptible for a one-unit increase in length, holding area and year fixed. With $\hat{\gamma}_{\text{Length}} = -0.0099$, the odds of being non-susceptible is 0.9901 ($= \exp(-0.0099)$) times of that for a fish one unit shorter. The negative coefficient means longer fish are less likely to be from the non-susceptible strain (i.e., longer fish are more likely susceptible to parasites).